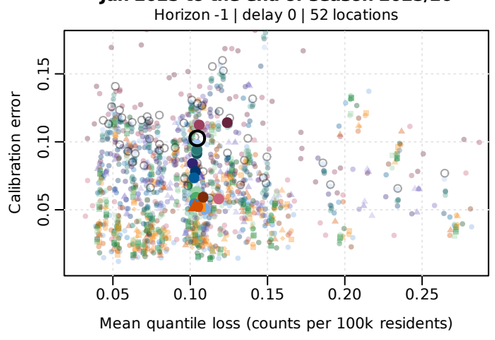
covid | all 52 locations (without US) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k reside

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo
Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.

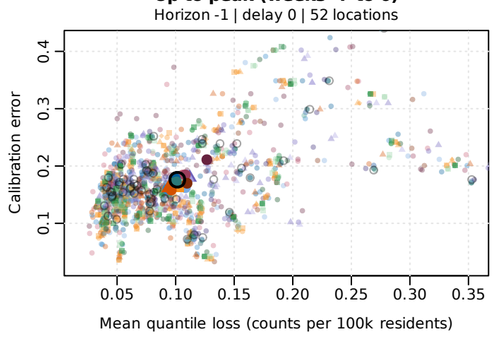
Horizon -1

Feedback delay 0

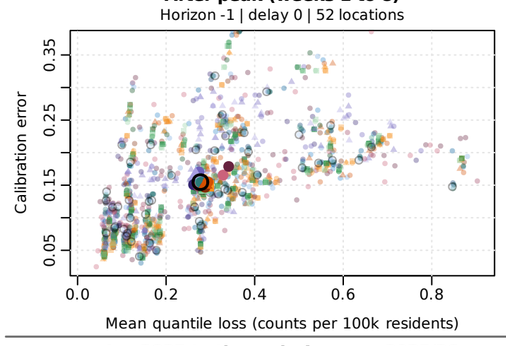
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 0 | 52 locations



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 52 locations

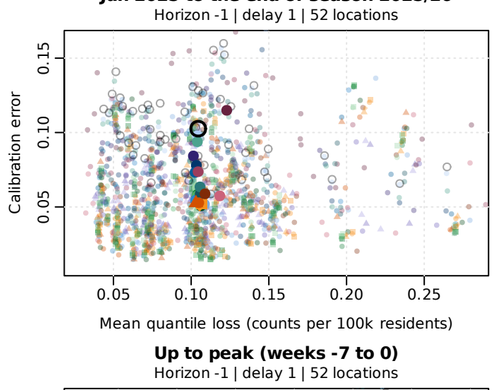


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 52 locations

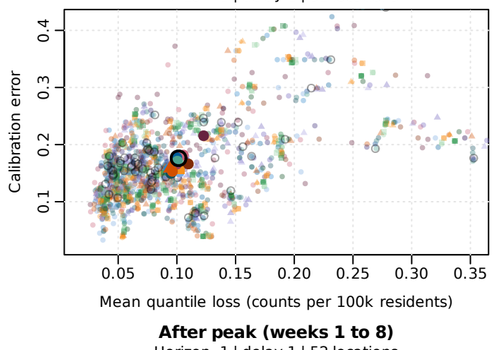


Feedback delay 1

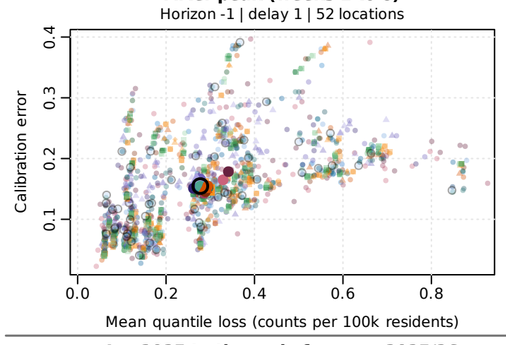
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 1 | 52 locations



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 52 locations

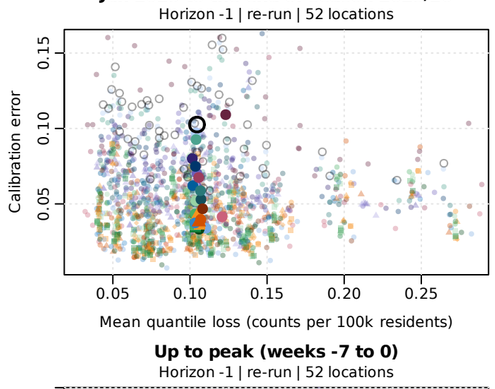


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 52 locations

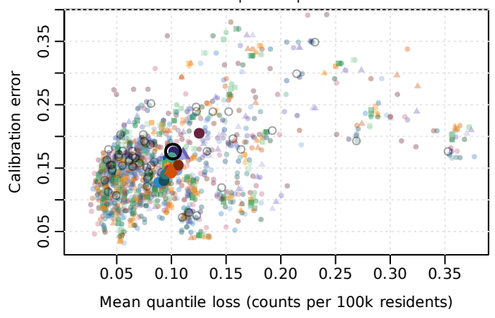


Feedback delay 0 + revisions (re-run)

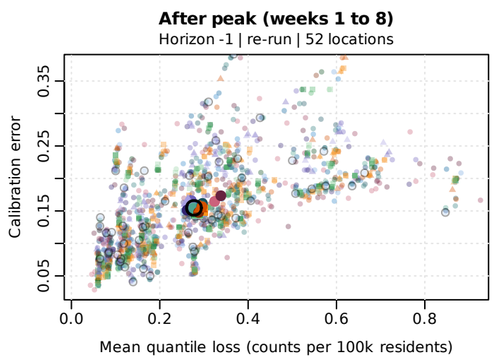
Jun 2025 to the end of season 2025/26
Horizon -1 | re-run | 52 locations



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 52 locations



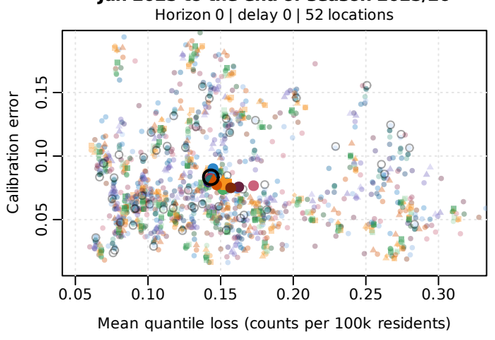
After peak (weeks 1 to 8)
Horizon -1 | re-run | 52 locations



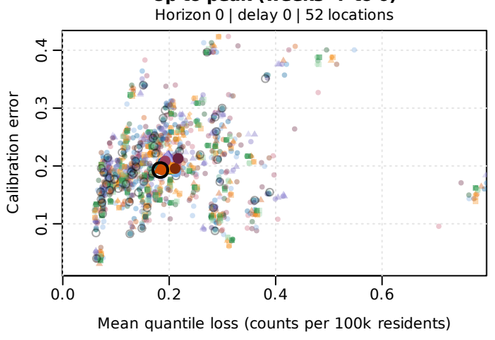
Horizon 0

Feedback delay 0

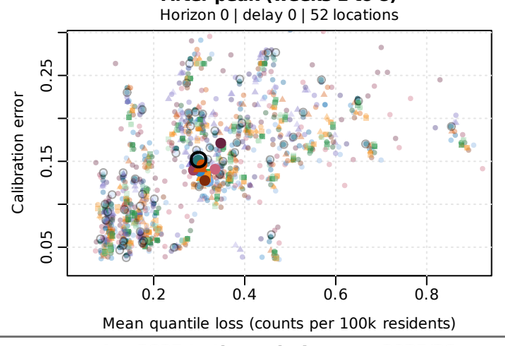
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 0 | 52 locations



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 52 locations

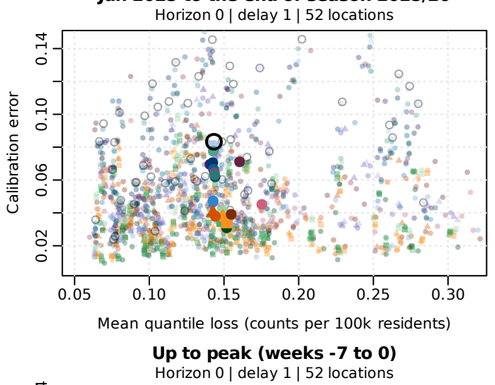


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 52 locations

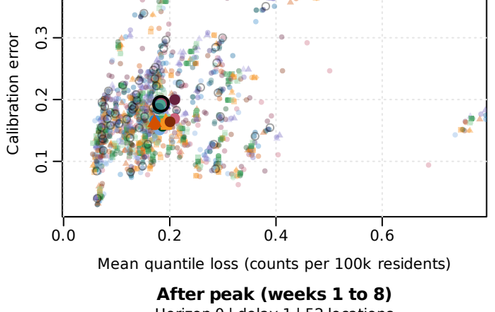


Feedback delay 1

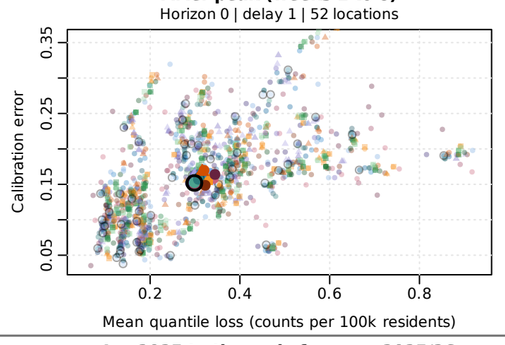
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 1 | 52 locations



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 52 locations

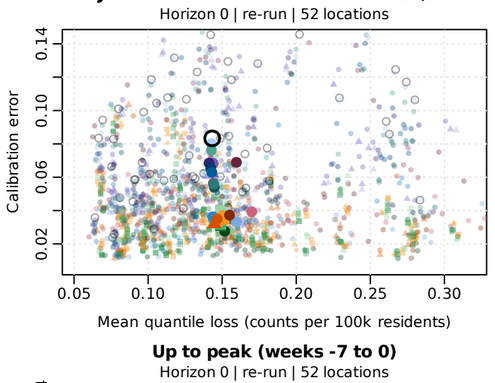


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 52 locations

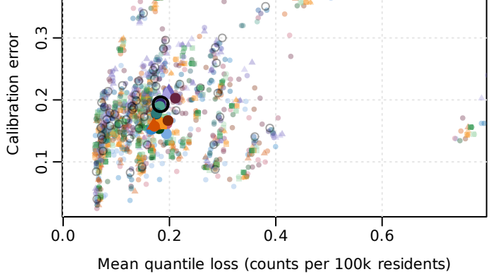


Feedback delay 0 + revisions (re-run)

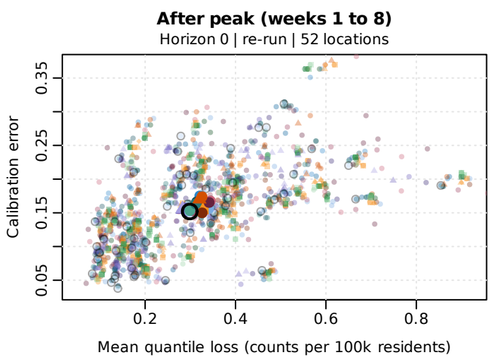
Jun 2025 to the end of season 2025/26
Horizon 0 | re-run | 52 locations



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 52 locations



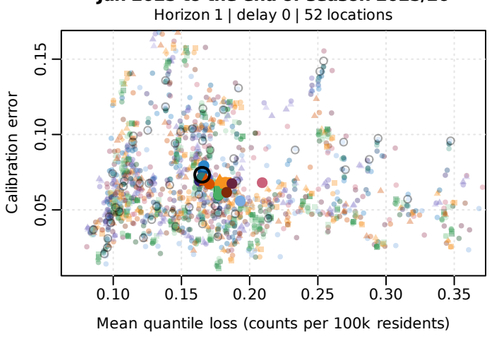
After peak (weeks 1 to 8)
Horizon 0 | re-run | 52 locations



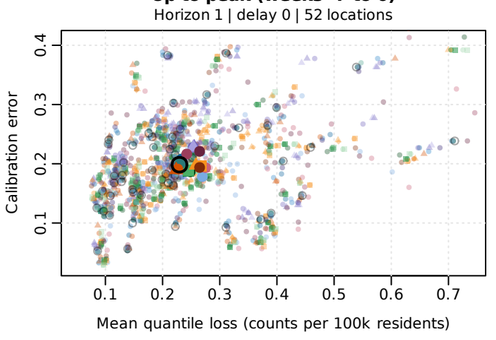
Horizon 1

Feedback delay 0

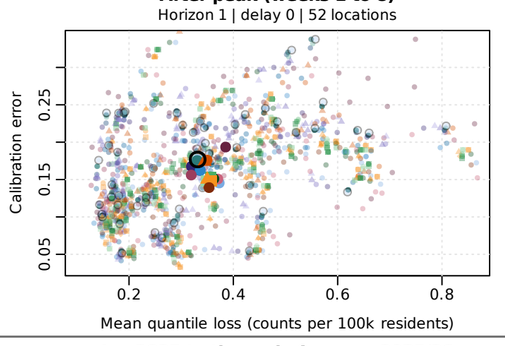
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 0 | 52 locations



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 52 locations

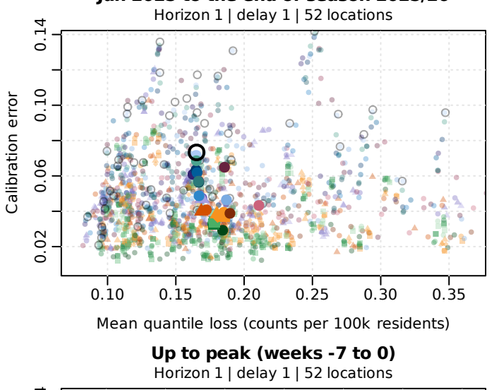


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 52 locations

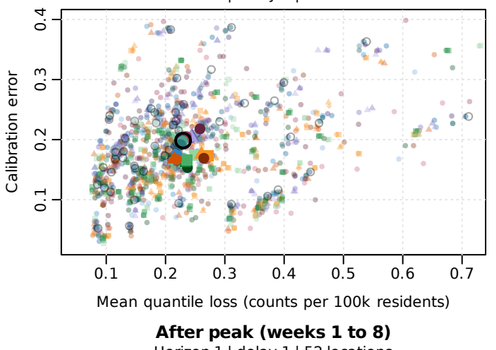


Feedback delay 1

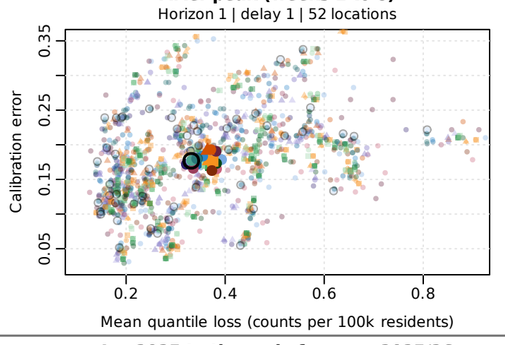
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 1 | 52 locations



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 52 locations

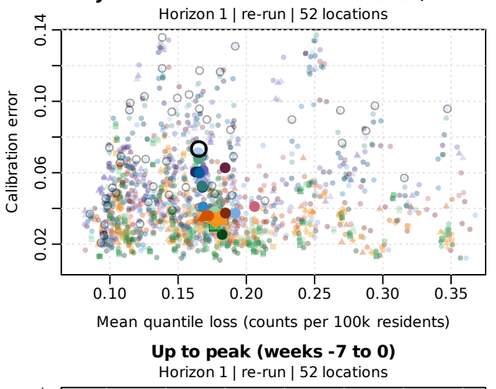


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 52 locations

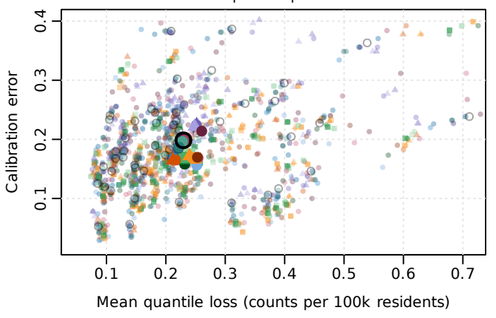


Feedback delay 0 + revisions (re-run)

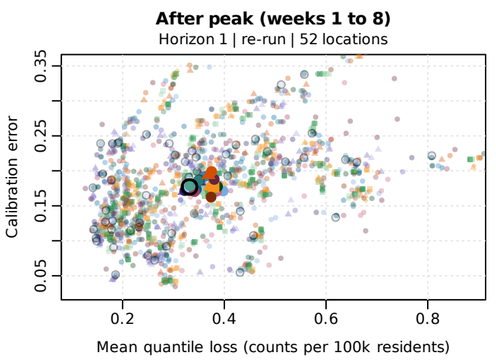
Jun 2025 to the end of season 2025/26
Horizon 1 | re-run | 52 locations



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 52 locations



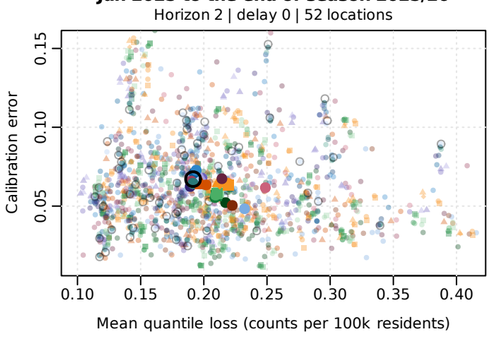
After peak (weeks 1 to 8)
Horizon 1 | re-run | 52 locations



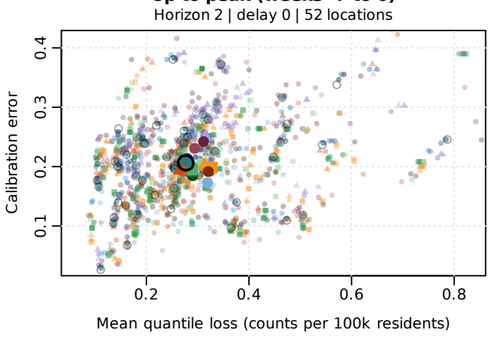
Horizon 2

Feedback delay 0

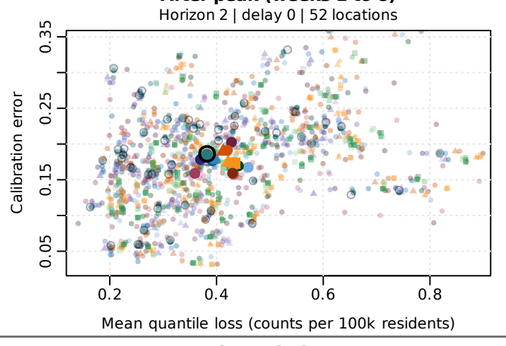
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 0 | 52 locations



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 52 locations

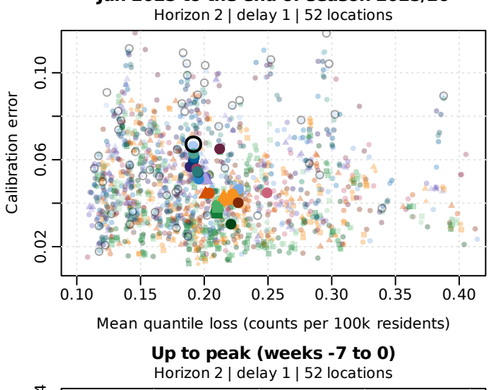


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 52 locations

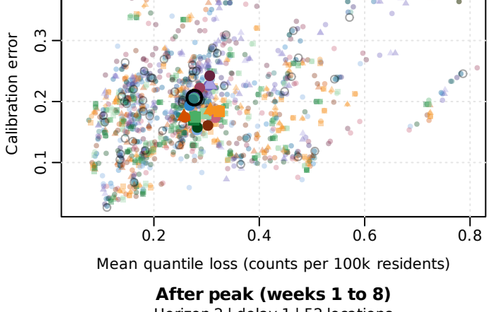


Feedback delay 1

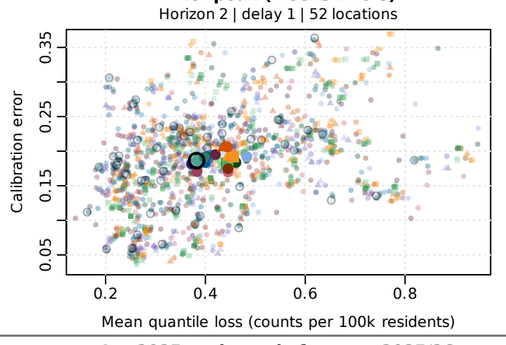
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 1 | 52 locations



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 52 locations

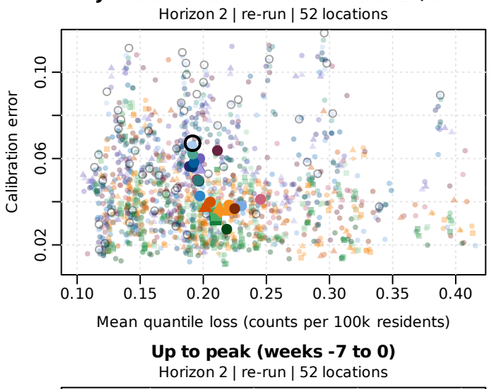


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 52 locations

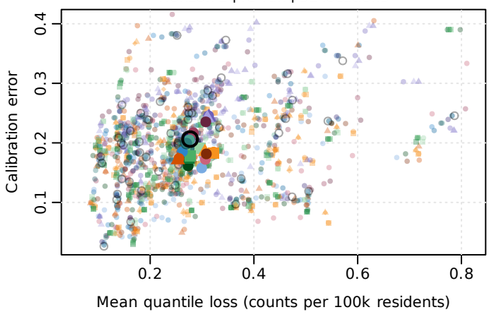


Feedback delay 0 + revisions (re-run)

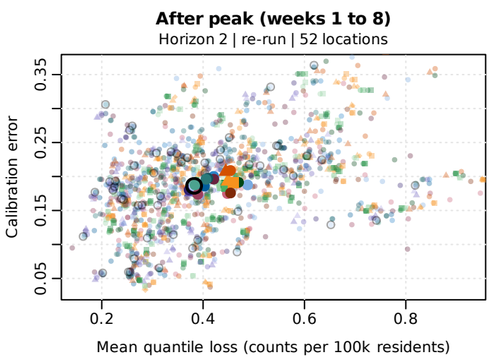
Jun 2025 to the end of season 2025/26
Horizon 2 | re-run | 52 locations



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 52 locations



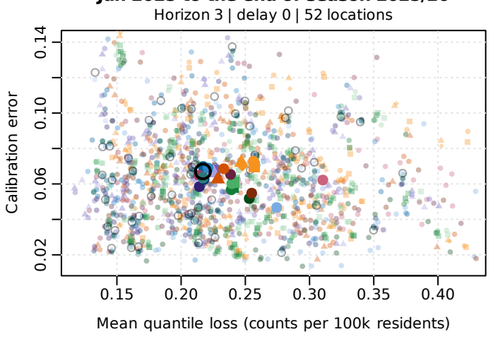
After peak (weeks 1 to 8)
Horizon 2 | re-run | 52 locations



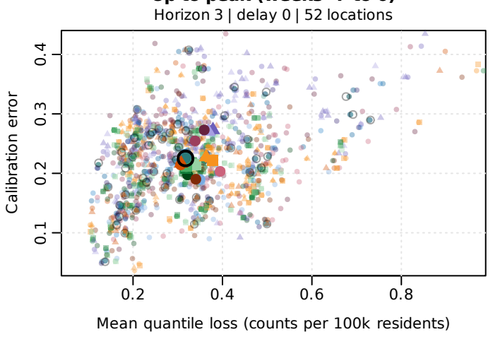
Horizon 3

Feedback delay 0

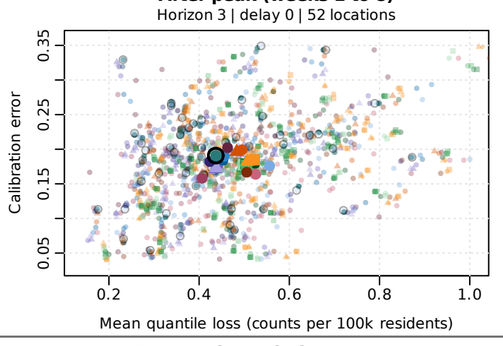
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 0 | 52 locations



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 52 locations

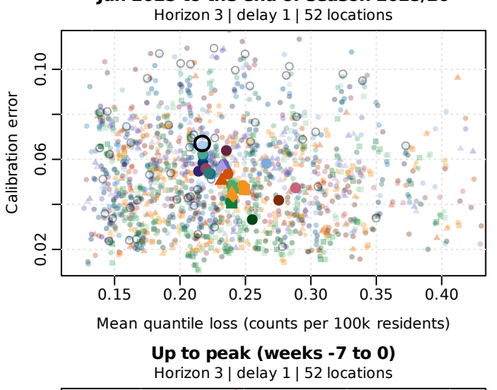


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 52 locations

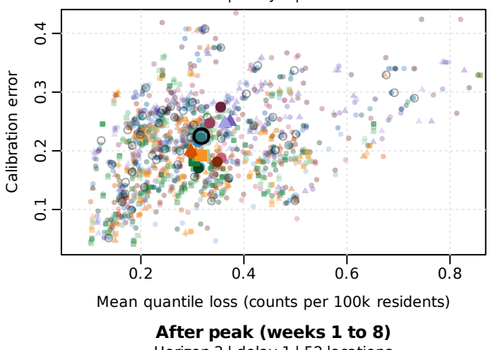


Feedback delay 1

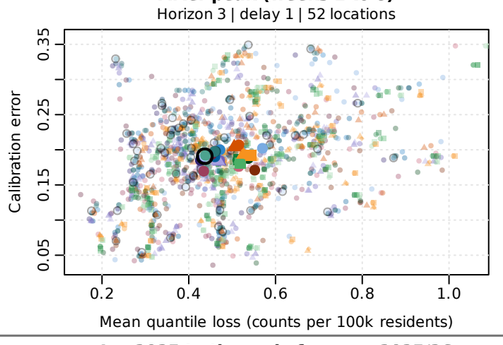
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 1 | 52 locations



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 52 locations

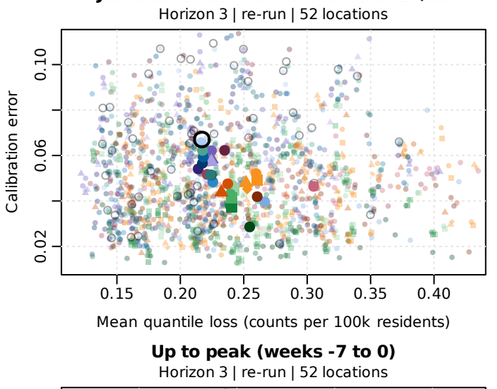


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 52 locations

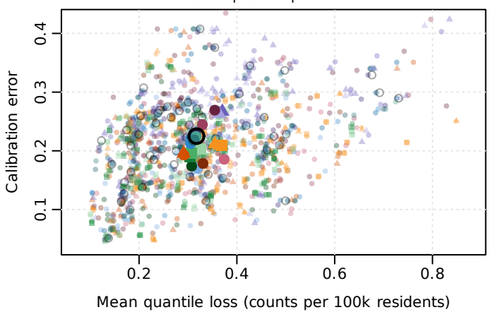


Feedback delay 0 + revisions (re-run)

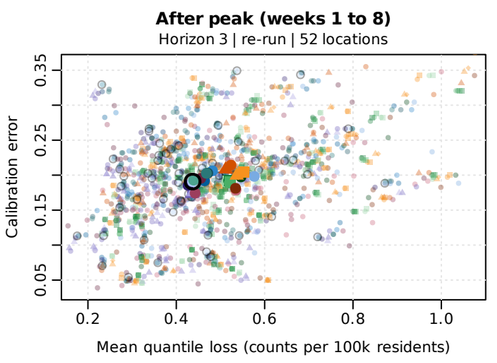
Jun 2025 to the end of season 2025/26
Horizon 3 | re-run | 52 locations



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 52 locations



After peak (weeks 1 to 8)
Horizon 3 | re-run | 52 locations



trackers
- l
- ultiQT (eta=0.001*S0)
- REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- flu prod (sqrt per 100k, eta=0.018)
- slope rate (min m=0.05)
- oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=4)
- Rate grid: learned + sharing, common weights (H=4)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Diverse experts
- Diverse experts: equal (slow m=0.001)
- Diverse experts: learned + sharing (slow m=0.001, H=8)
- Diverse experts: learned + sharing, common weights (slow m=0.001)
- Diverse experts: soft switching (strength=1)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.00
- Studentized median (base 80% width, normal start, et
- Median, constant sigma (RMS of 5, normal start, eta=

Studentized full base
- Studentized full base (base 80% width, eta=0.003)
- Studentized full base (level scales, eta=0.003)
- Studentized full base (level RMS of 5, eta=0.003)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback protocol latest-at-use-v2: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback protocol latest-at-use-v2: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | California (06) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo

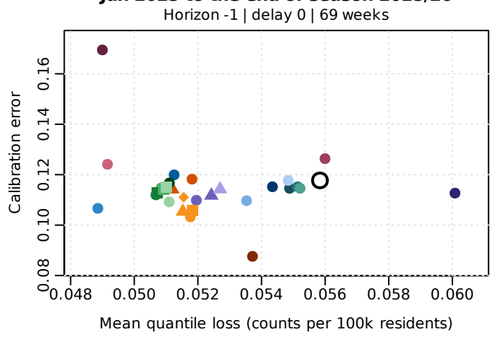
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback protocol latest-at-use-v2: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | Texas (48) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

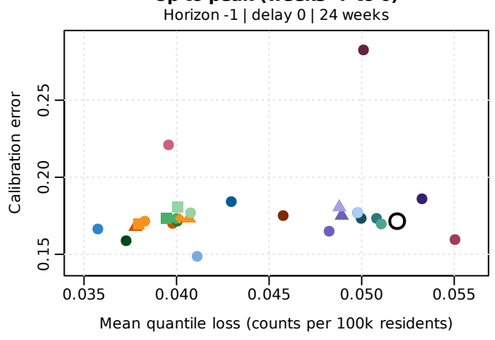
Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo

Horizon -1

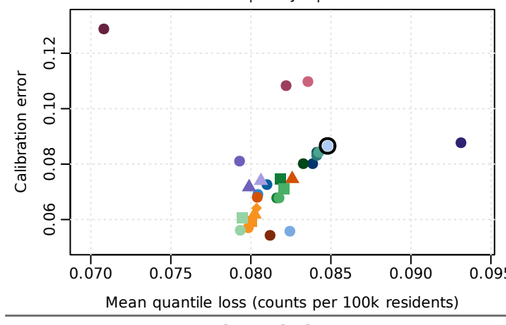
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 0 | 69 weeks



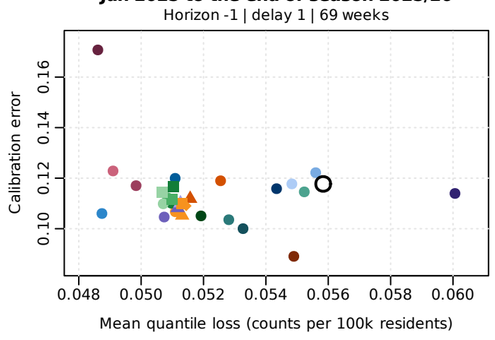
Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 24 weeks



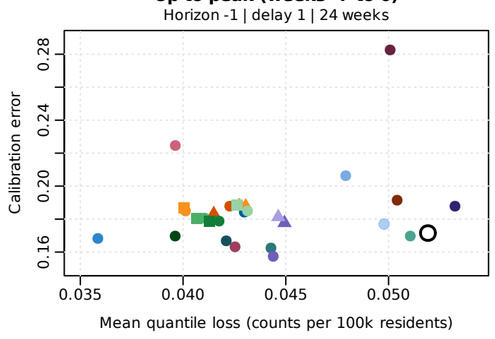
After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 19 weeks



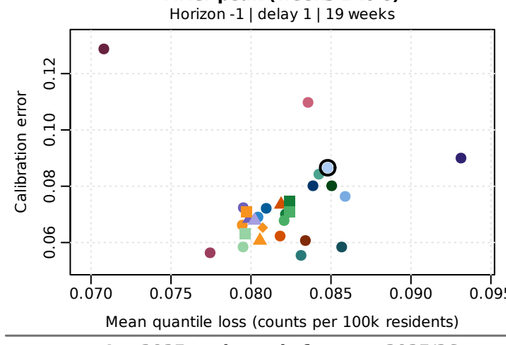
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 1 | 69 weeks



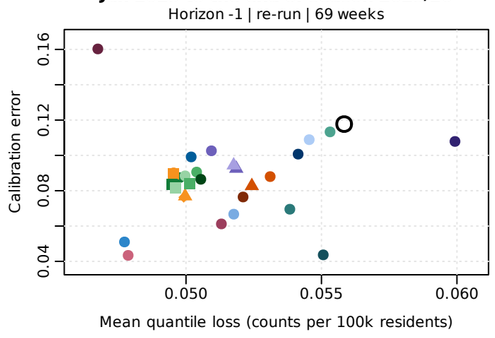
Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 24 weeks



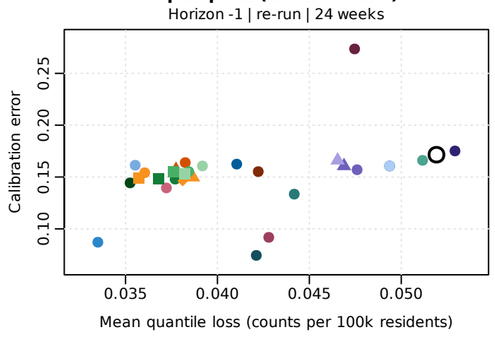
After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 19 weeks



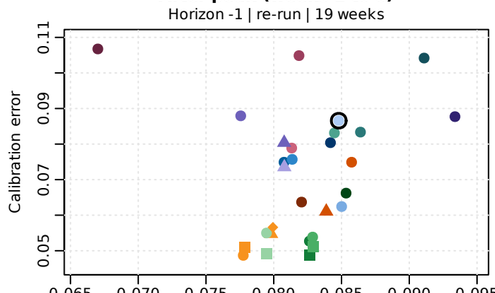
Jun 2025 to the end of season 2025/26
Horizon -1 | re-run | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 24 weeks

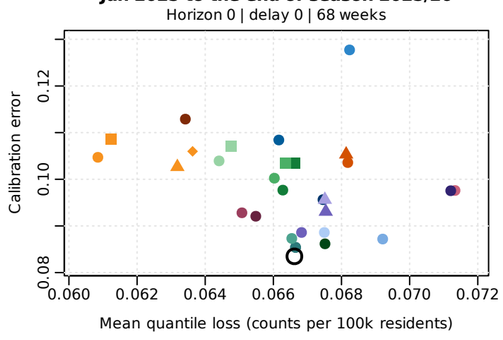


After peak (weeks 1 to 8)
Horizon -1 | re-run | 19 weeks

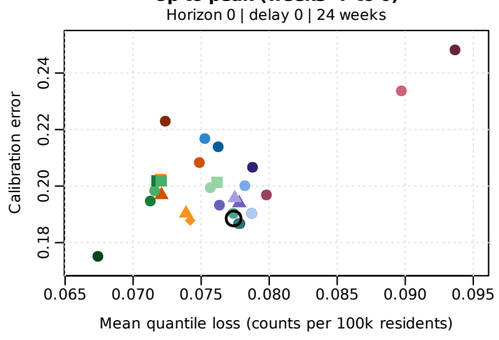


Horizon 0

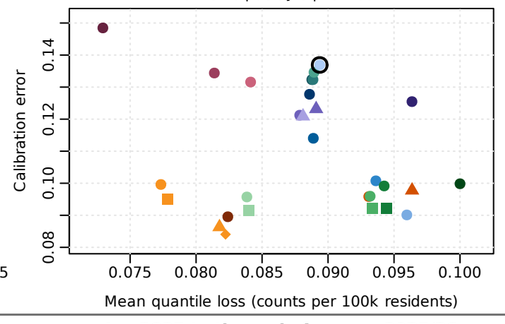
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 0 | 68 weeks



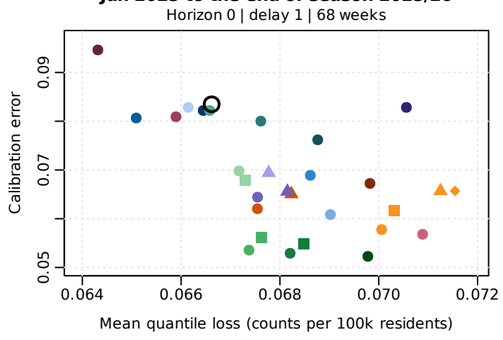
Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 24 weeks



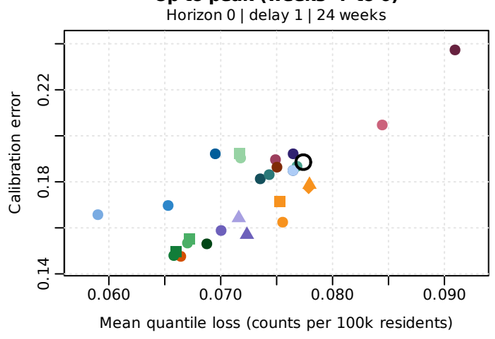
After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 19 weeks



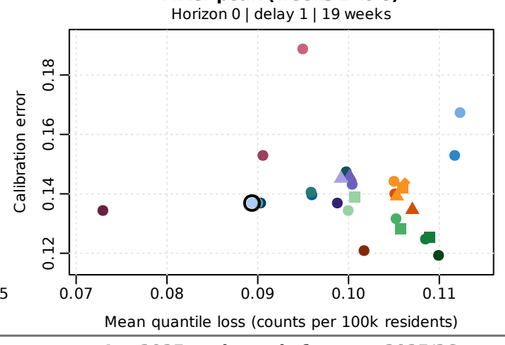
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 1 | 68 weeks



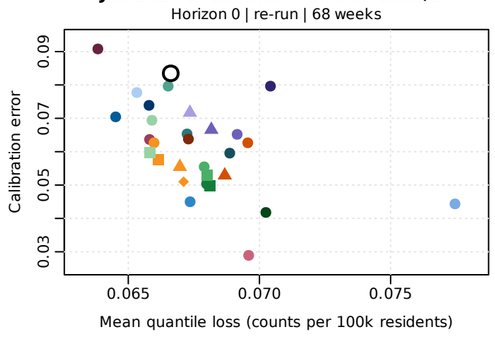
Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 24 weeks



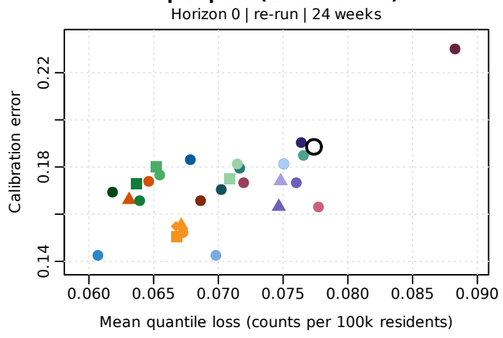
After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 19 weeks



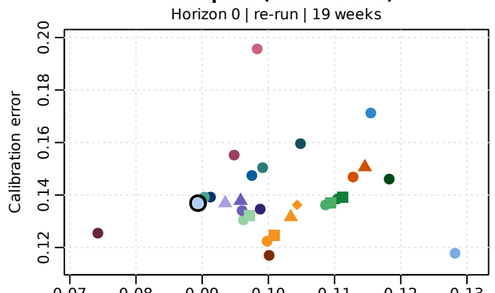
Jun 2025 to the end of season 2025/26
Horizon 0 | re-run | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 24 weeks

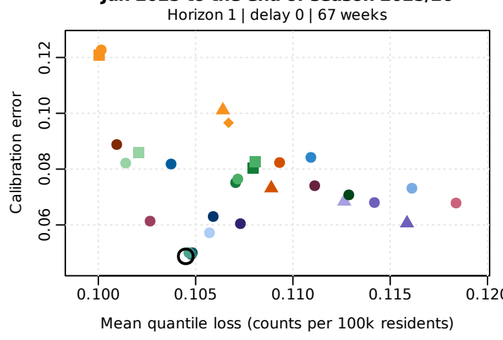


After peak (weeks 1 to 8)
Horizon 0 | re-run | 19 weeks

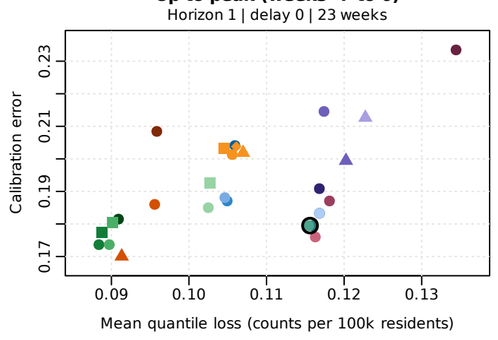


Horizon 1

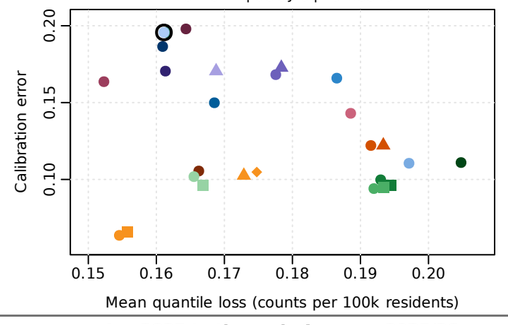
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 0 | 67 weeks



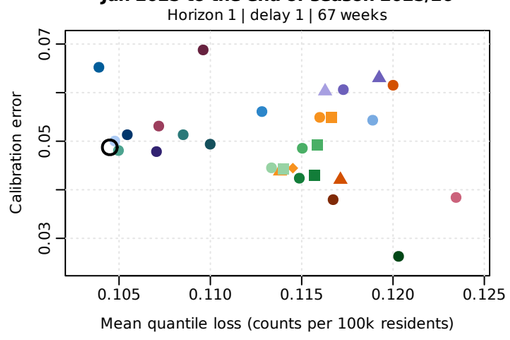
Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 23 weeks



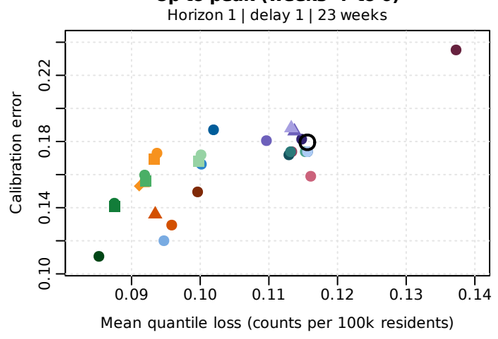
After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 19 weeks



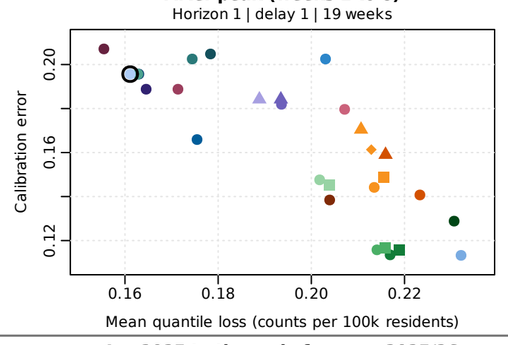
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 1 | 67 weeks



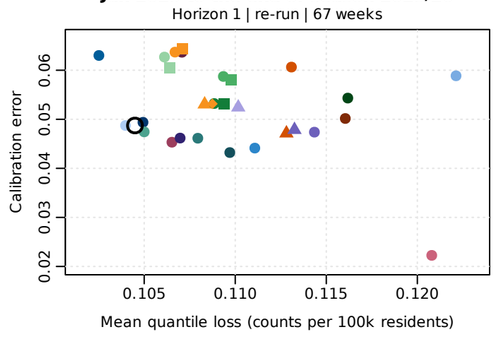
Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 23 weeks



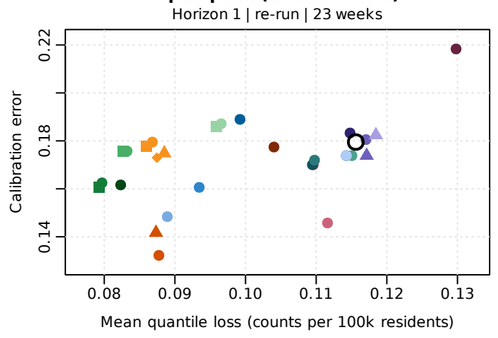
After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 19 weeks



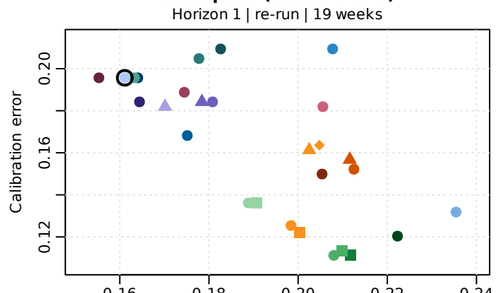
Jun 2025 to the end of season 2025/26
Horizon 1 | re-run | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 23 weeks

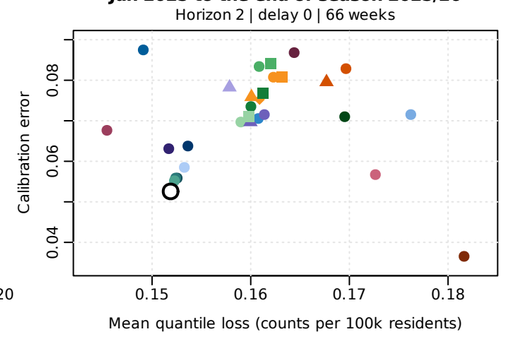


After peak (weeks 1 to 8)
Horizon 1 | re-run | 19 weeks

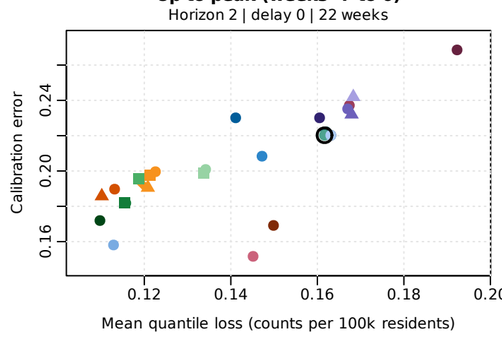


Horizon 2

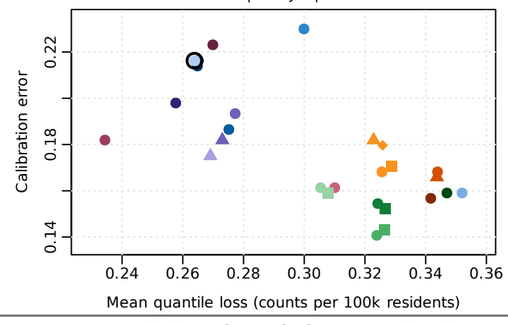
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 0 | 66 weeks



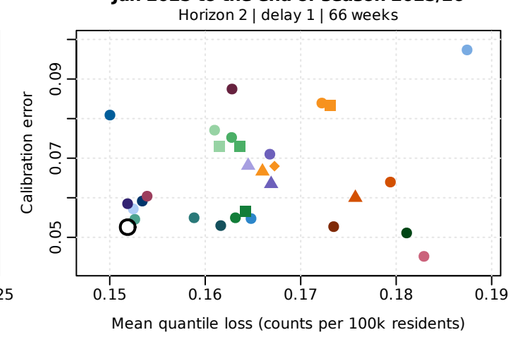
Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 22 weeks



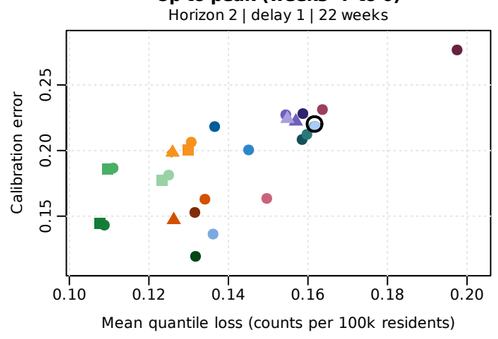
After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 19 weeks



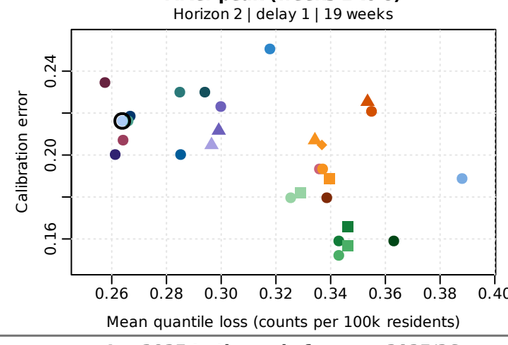
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 1 | 66 weeks



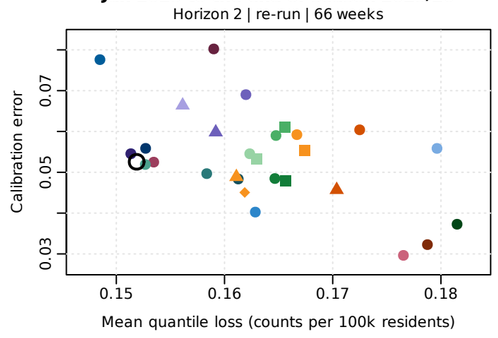
Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 22 weeks



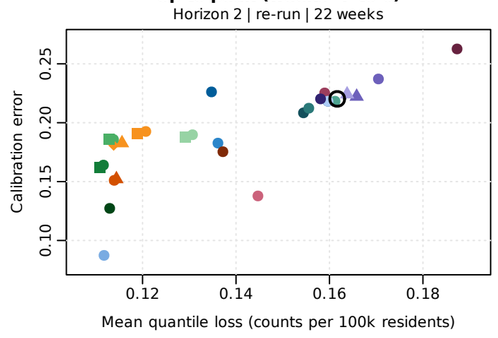
After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 19 weeks



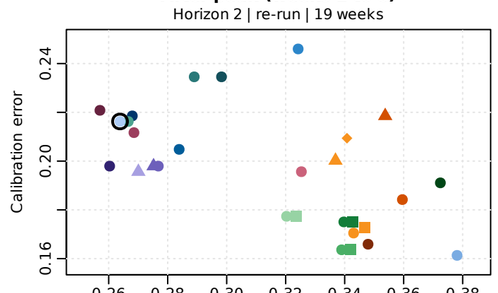
Jun 2025 to the end of season 2025/26
Horizon 2 | re-run | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 22 weeks

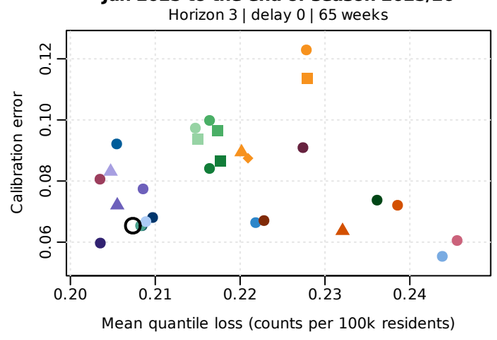


After peak (weeks 1 to 8)
Horizon 2 | re-run | 19 weeks

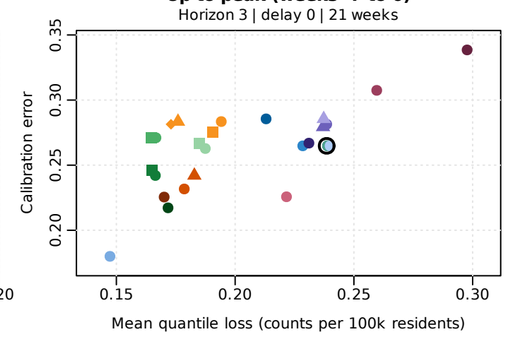


Horizon 3

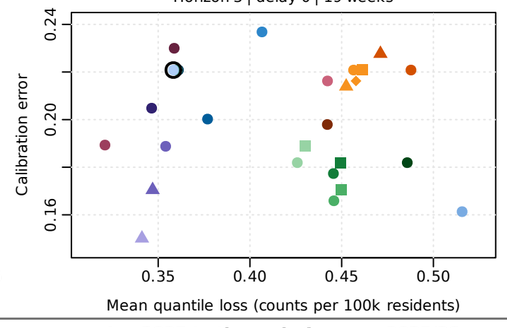
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 0 | 65 weeks



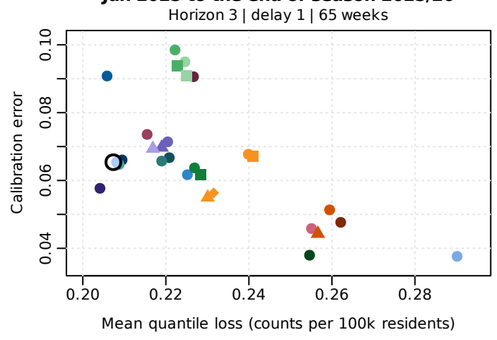
Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 21 weeks



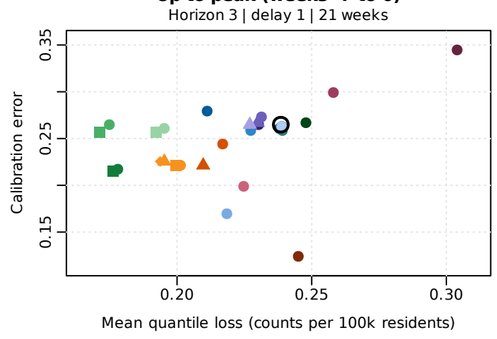
After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 19 weeks



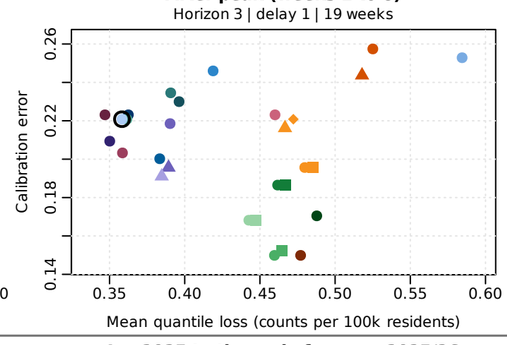
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 1 | 65 weeks



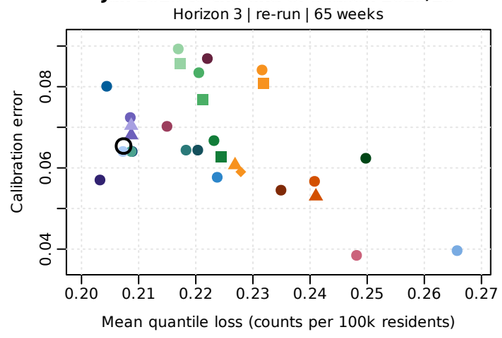
Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 21 weeks



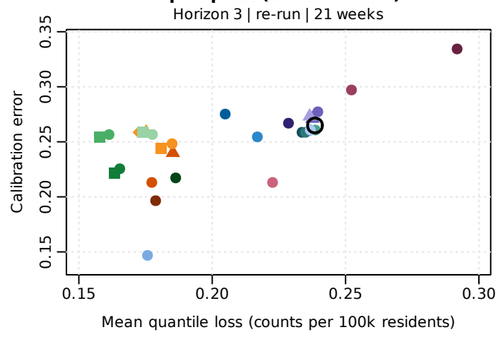
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 19 weeks



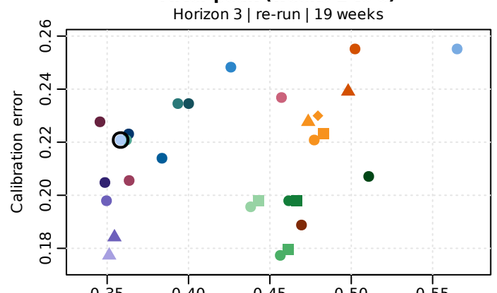
Jun 2025 to the end of season 2025/26
Horizon 3 | re-run | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 21 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 19 weeks



Feedback delay 0

Feedback delay 1

Feedback delay 0 + revisions (re-run)

trackers
- l
- ultiQT (eta=0.001*S0)
- REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- flu prod (sqrt per 100k, eta=0.018)
- slope rate (min m=0.05)
- oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=4)
- Rate grid: learned + sharing, common weights (H=4)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Diverse experts
- Diverse experts: equal (slow m=0.001)
- Diverse experts: learned + sharing (slow m=0.001, H=8)
- Diverse experts: learned + sharing, common weights (slow m=0.001)
- Diverse experts: soft switching (strength=1)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.00
- Studentized median (base 80% width, normal start, et
- Median, constant sigma (RMS of 5, normal start, eta=

Studentized full base
- Studentized full base (base 80% width, eta=0.003)
- Studentized full base (level scales, eta=0.003)
- Studentized full base (level RMS of 5, eta=0.003)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback protocol latest-at-use-v2: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | New York (36) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo

trackers
- l
- ultiQT (eta=0.001*S0)
- REF-op (sqrt, eta=0.03*Q90 of 20, slow 0.003, leak 0.1)
- flu prod (sqrt per 100k, eta=0.018)
- slope rate (min m=0.05)
- oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=4)
- Rate grid: learned + sharing, common weights (H=4)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Diverse experts
- Diverse experts: equal (slow m=0.001)
- Diverse experts: learned + sharing (slow m=0.001, H=8)
- Diverse experts: learned + sharing, common weights (slow m=0.001)
- Diverse experts: soft switching (strength=1)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.00
- Studentized median (base 80% width, normal start, et
- Median, constant sigma (RMS of 5, normal start, eta=

Studentized full base
- Studentized full base (base 80% width, eta=0.003)
- Studentized full base (level scales, eta=0.003)
- Studentized full base (level RMS of 5, eta=0.003)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback protocol latest-at-use-v2: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | Pennsylvania (42) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (protocol latest-at-use-v2; delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizo

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Feedback protocol latest-at-use-v2: Wednesday issue cutoffs (actual generation dates where known); S0 and the weighting loss scale from the first 20 released rows in hindsight, frozen for the re-runs.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).